

smooth to a radius greater than or equal to half the branch thickness. Final grinding marks should be transverse to the weld axis and the entire finished profile should pass magnetic particle inspection.

5.5 STRESS CONCENTRATION FACTORS

The X and X' curves should be used with hot spot stress ranges based on suitable stress concentration factors. Stress concentration factors may be derived from finite element analyses, model tests or empirical equations based on such methods.

For joints not meeting the requirements of Section 4.3.1, e.g., connections in which load transfer is accomplished by overlap (Section 4.3.2), or by gusset plates, ring stiffeners, etc. (Section 4.3.5), a minimum stress concentration factor of 6.0 should be used in the brace member, in lieu of additional analysis. Where the chord and/or other joint reinforcement are not designed to develop the full static capacity of the members joined, these elements should also be checked separately.

6 Foundation Design

The recommended criteria of Section 6.1 through Section 6.11 are devoted to pile foundations, and more specifically to steel cylindrical (pipe) pile foundations. The recommended criteria of Section 6.12 through Section 6.17 are devoted to shallow foundations.

6.1 GENERAL

The foundation should be designed to carry static, cyclic and transient loads without excessive deformations or vibrations in the platform. Special attention should be given to the effects of cyclic and transient loading on the strength of the supporting soils as well as on the structural response of piles. Guidance provided in Sections 6.3, 6.4, and 6.5 is based upon static, monotonic loadings. Furthermore, this guidance does not necessarily apply to so called problem soils such as carbonate material or volcanic sands or highly sensitive clays. The possibility of movement of the seafloor against the foundation members should be investigated and the forces caused by such movements, if anticipated, should be considered in the design.

6.2 PILE FOUNDATIONS

Types of pile foundations used to support offshore structures are as follows:

6.2.1 Driven Piles

Open ended piles are commonly used in foundations for offshore platforms. These piles are usually driven into the sea-floor with impact hammers which use steam, diesel fuel,

or hydraulic power as the source of energy. The pile wall thickness should be adequate to resist axial and lateral loads as well as the stresses during pile driving. It is possible to predict approximately the stresses during pile driving using the principles of one-dimensional elastic stress wave transmission by carefully selecting the parameters that govern the behavior of soil, pile, cushions, capblock and hammer. For a more detailed study of these principles, refer to E.A.L. Smith's paper, "Pile Driving Analysis by the Wave Equation," Transactions ASCE, Vol. 127, 1962, Part 1, Paper No. 3306, pp. 1145–1193. The above approach may also be used to optimize the pile hammer cushion and capblock with the aid of computer analyses (commonly known as the Wave Equation Analyses). The design penetration of driven piles should be determined in accordance with the principles outlined in Sections 6.3 through 6.7 and 6.9 rather than upon any correlation of pile capacity with the number of blows required to drive the pile a certain distance into the seafloor.

When a pile refuses before it reaches design penetration, one or more of the following actions can be taken:

- a. Review of hammer performance. A review of all aspects of hammer performance, possibly with the aid of hammer and pile head instrumentation, may identify problems which can be solved by improved hammer operation and maintenance, or by the use of a more powerful hammer.
- b. Reevaluation of design penetration. Reconsideration of loads, deformations and required capacities, of both individual piles and other foundation elements, and the foundation as a whole, may identify reserve capacity available. An interpretation of driving records in conjunction with instrumentation mentioned above may allow design soil parameters or stratification to be revised and pile capacity to be increased.
- c. Modifications to piling procedures, usually the last course of action, may include one of the following:
 - **Plug Removal.** The soil plug inside the pile is removed by jetting and air lifting or by drilling to reduce pile driving resistance. If plug removal results in inadequate pile capacities, the removed soil plug should be replaced by a gravel grout or concrete plug having sufficient load-carrying capacity to replace that of the removed soil plug. Attention should be paid to plug/pile load transfer characteristics. Plug removal may not be effective in some circumstances particularly in cohesive soils.
 - **Soil Removal Below Pile Tip.** Soil below the pile tip is removed either by drilling an undersized hole or jetting equipment is lowered through the pile which acts as the casing pipe for the operation. The effect on pile capacity of drilling an undersized hole is unpredictable unless there has been previous experience under similar conditions. Jetting below the pile tip should in general be avoided because of the unpredictability of the results.

- **Two-State Driven Piles.** A first stage or outer pile is driven to a predetermined depth, the soil plug is removed, and a second stage or inner pile is driven inside the first stage pile. The annulus between the two piles is grouted to permit load transfer and develop composite action.
- Drilled and grouted insert piles as described in 6.2.2(b) below.

6.2.2 Drilled and Grouted Piles

Drilled and grouted piles can be used in soils which will hold an open hole with or without drilling mud. Load transfer between grout and pile should be designed in accordance with Sections 7.4.2, 7.4.3, and 7.4.4. There are two types of drilled and grouted piles, as follows:

- a. **Single-Stage.** For the single-staged, drilled and grouted pile, an oversized hole is drilled to the required penetration, a pile is lowered into the hole and the annulus between the pile and the soil is grouted. This type pile can be installed only in soils which will hold an open hole to the surface. As an alternative method, the pile with expendable cutting tools attached to the tip can be used as part of the drill stem to avoid the time required to remove the drill bit and insert a pile.
- b. **Two-Stage.** The two-staged, drilled and grouted pile consists of two concentrically placed piles grouted to become a composite section. A pile is driven to a penetration which has been determined to be achievable with the available equipment and below which an open hole can be maintained. This outer pile becomes the casing for the next operation which is to drill through it to the required penetration for the inner or “insert” pile. The insert pile is then lowered into the drilled hole and the annuli between the insert pile and the soil an between the two piles are grouted. Under certain oil conditions, the drilled hole is stopped above required penetration, and the insert pile is driven to required penetration. The diameter of the drilled hole should be at least 6 inches (150 mm) larger than the pile diameter.

6.2.3 Belled Piles

Bells may be constructed at the tip of piles to give increased bearing and uplift capacity through direct bearing on the soil. Drilling of the bell is carried out through the pile by under-reaming with an expander tool. A pilot hole may be drilled below the bell to act as a sump for unrecoverable cuttings. The bell and pile are filled with concrete to a height sufficient to develop necessary load transfer between the bell and the pile. Bells are connected to the pile to transfer full uplift and bearing loads using steel reinforcing such as structural members with adequate shear lugs, deformed reinforcement bars or pre-stressed tendons. Load transfer into the concrete should be designed in accordance with ACI 318. The steel reinforcing should be enclosed for their full length below the pile with

spiral reinforcement meeting the requirements of ACI 318. Load transfer between the concrete and the pile should be designed in accordance with Sections 7.4.2, 7.4.3, and 7.4.4.

6.3 PILE DESIGN

6.3.1 Foundation Size

When sizing a pile foundation, the following items should be considered: diameter, penetration, wall thickness, type of tip, spacing, number of piles, geometry, location, mudline restraint, material strength, installation method, and other parameters as may be considered appropriate.

6.3.2 Foundation Response

A number of different analysis procedures may be utilized to determine the requirements of a foundation. At a minimum, the procedure used should properly stimulate the non-linear response behavior of the soil and assure load-deflection compatibility between the structure and the pile-soil system.

6.3.3 Deflections and Rotations

Deflections and rotations of individual piles and the total foundation system should be checked at all critical locations which may include pile tops, points of contraflexure, mudline, etc. Deflections and rotations should not exceed serviceability limits which would render the structure inadequate for its intended function.

6.3.4 Pile Penetration

The design pile penetration should be sufficient to develop adequate capacity to resist the maximum computed axial bearing and pullout loads with an appropriate factor of safety. The ultimate pile capacities can be computed in accordance with Sections 6.4 and 6.5 or by other methods which are supported by reliable comprehensive data. The allowable pile capacities are determined by dividing the ultimate pile capacities by appropriate factors of safety which should not be less than the following values:

Load Condition	Factors of Safety
1. Design environmental conditions with appropriate drilling loads	1.5
2. Operating environmental conditions during drilling operations	2.0
3. Design environmental conditions with appropriate producing loads	1.5
4. Operating environmental conditions during producing operations	2.0
5. Design environmental conditions with minimum loads (for pullout)	1.5

6.3.5 Alternative Design Methods

The provisions of this recommended practice for sizing the foundation pile are based on an allowable stress (working stress) method except for pile penetration per Section 6.3.4. In this method, the foundation piles should conform to the requirements of Sections 3.2 and 6.10 in addition to the provisions of Section 6.3. Any alternative method supported by sound engineering methods and empirical evidence may also be utilized. Such alternative methods include the limit state design approach or ultimate strength design of the total foundation system.

6.3.6 Scour

Seabed scour affects both lateral and axial pile performance and capacity. Scour prediction remains an uncertain art. Sediment transport studies may assist in defining scour design criteria but local experience is the best guide. The uncertainty on design criteria should be handled by robust design, or by an operating strategy of monitoring and remediation as needed. Typical remediation experience is documented in “Erosion Protection of Production Structures,” by Posey, C.J., and Sybert, J.H., Proc. 9th Conv. I.A.H.R., Dobrovnik, 1961, pp. 1157-1162, and “Scour Repair Methods in the Southern North Sea,” by Angus, N.M., and Moore, R.L., OTC 4410, May 1982. Scour design criteria will usually be a combination of local and global scour.

6.4 PILE CAPACITY FOR AXIAL BEARING LOADS

6.4.1 Ultimate Bearing Capacity

The ultimate bearing capacity of piles, including belled piles, Q_d should be determined by the equation:

$$Q_d = Q_f + Q_p = fA_s + qA_p \quad (6.4.1-1)$$

where

Q_f = skin friction resistance, lb (kN),

Q_p = total end bearing, lb (kN),

f = unit skin friction capacity, lb/ft² (kPa),

A_s = side surface area of pile, ft² (m²),

q = unit end bearing capacity, lb/ft² (kPa),

A_p = gross end area of pile, ft² (m²).

Total end bearing, Q_p , should not exceed the capacity of the internal plug. In computing pile loading and capacity the weight of the pile-soil plug system and hydrostatic uplift should be considered.

In determining the load capacity of a pile, consideration should be given to the relative deformations between the soil

and the pile as well as the compressibility of the soil pile system. Eq. 6.4.1-1 assumes that the maximum skin friction along the pile and the maximum end bearing are mobilized simultaneously. However, the ultimate skin friction increments along the pile are not necessarily directly additive, nor is the ultimate end bearing necessarily additive to the ultimate skin friction. In some circumstances this effect may result in the capacity being less than that given by Eq. 6.4.1-1. In such cases a more explicit consideration of axial pile performance effects on pile capacity may be warranted. For additional discussion of these effects refer to Section 6.6 and ASCE *Journal of the Soil Mechanics and Foundations Division for Load Transfer for Axially Loaded Piles in Clay*, by H.M. Coyle and L.C. Reese, Vol. 92, No. 1052, March 1966, Murff, J.D., “Pile Capacity in a Softening Soil,” *International Journal for Numerical and Analytical Methods in Geomechanics* (1980), Vol. 4, No. 2, pp. 185–189, and Randolph, H.F., “Design Considerations for Offshore Piles,” *Geotechnical Practice in Offshore Engineering*, ASCE, Austin 1983, pp. 422–439.

The foundation configurations should be based on those that experience has shown can be installed consistently, practically and economically under similar conditions with the pile size and installation equipment being used. Alternatives for possible remedial action in the event design objectives cannot be obtained during installation should also be investigated and defined prior to construction.

For the pile-bell system, the factors of safety should be those given in Section 6.3.4. The allowable skin friction values on the pile section should be those given in this section and in Section 6.5. Skin friction on the upper bell surface and possibly above the bell on the pile should be discounted in computing skin friction resistance, Q_f . The end bearing area of a pilot hole, if drilled, should be discounted in computing total bearing area of the bell.

6.4.2 Skin Friction and End Bearing in Cohesive Soils

For pipe piles in cohesive soils, the shaft friction, f , in lb/ft² (kPa) at any point along the pile may be calculated by the equation.

$$f = \alpha c \quad (6.4.2-1)$$

where

α = a dimensionless factor,

c = undrained shear strength of the soil at the point in question.

The factor, α , can be computed by the equations:

$$\alpha = 0.5 \psi^{-0.5} \psi \leq 1.0 \quad (6.4.2-2)$$

$$\alpha = 0.5 \psi^{-0.25} \psi > 1.0$$

with the constraint that, $\alpha \leq 1.0$,

where

$\psi = c/p'_o$ for the point in question,

$p'_o =$ effective overburden pressure at the point in question lb/ft² (kPa).

A discussion of appropriate methods for determining the undrained shear strength, c , and effective overburden pressure, p'_o , including the effects of various sampling and testing procedures is included in the commentary. For underconsolidated clays (clays with excess pore pressures undergoing active consolidation), α , can usually be taken as 1.0. Due to the lack of pile load tests in soils having c/p'_o ratios greater than three, equation 6.4.2-2 should be applied with some engineering judgment for high c/p'_o values. Similar judgment should be applied for deep penetrating piles in soils with high undrained shear strength, c , where the computed shaft frictions, f , using equation 6.4.2-1 above, are generally higher than previously specified in RP 2A.

For very long piles some reduction in capacity may be warranted, particularly where the shaft friction may degrade to some lesser residual value on continued displacement. This effect is discussed in more detail in the commentary.

Alternative means of determining pile capacity that are based on sound engineering principles and are consistent with industry experience are permissible. A more detailed discussion of alternate prediction methods is included in the commentary.

For piles end bearing in cohesive soils, the unit end bearing q , in lbs/ft² (kPa), may be computed by the equation

$$q = 9c \quad (6.4.2-3)$$

The shaft friction, f , acts on both the inside and outside of the pile. The total resistance is the sum of: the external shaft friction; the end bearing on the pile wall annulus; the total internal shaft friction or the end bearing of the plug, whichever is less. For piles considered to be plugged, the bearing pressure may be assumed to act over the entire cross section of the pile. For unplugged piles, the bearing pressure acts on the pile wall annulus only. Whether a pile is considered plugged or unplugged may be based on static calculations. For example, a pile could be driven in an unplugged condition but act plugged under static loading.

For piles driven in undersized drilled holes, piles jetted in place, or piles drilled and grouted in place the selection of shaft friction values should take into account the soil disturbance resulting from installation. In general f should not exceed values for driven piles; however, in some cases for drilled and grouted piles in overconsolidated clay, f may exceed these values. In determining f for drilled and grouted piles, the strength of the soil-grout interface, including poten-

tial effects of drilling mud, should be considered. A further check should be made of the allowable bond stress between the pile steel and the grout as recommended in Section 7.4.3. For further discussion refer to "State of the Art: Ultimate Axial Capacity of Grouted Piles" by Kraft and Lyons, OTC 2081, May, 1974.

In layered soils, shaft friction values, f , in the cohesive layers should be as given in Eq. (6.4.2-1). End bearing values for piles tipped in cohesive layers with adjacent weaker layers may be as given in Eq. (6.4.2-3), assuming that the pile achieves penetration of two to three diameters or more into the layer in question and the tip is approximately three diameters above the bottom of the layer to preclude punch through. Where these distances are not achieved, some modification in the end bearing resistance may be necessary. Where adjacent layers are of comparable strength to the layer of interest, the proximity of the pile tip to the interface is not a concern.

6.4.3 Shaft Friction and End Bearing in Cohesionless Soils

For pipe piles in cohesionless soils, the shaft friction, f , in lb/ft² (kPa) may be calculated by the equation:

$$f = Kp_o \tan \delta \quad (6.4.3-1)$$

where

$K =$ coefficient of lateral earth pressure (ratio of horizontal to vertical normal effective stress),

$p_o =$ effective overburden pressure lb/ft² (kPa) at the point in question,

$\delta =$ friction angle between the soil and pile wall.

For open-ended pipe piles driven unplugged, it is usually appropriate to assume K as 0.8 for both tension and compression loadings. Values of K for full displacement piles (plugged or closed end) may be assumed to be 1.0. Table 6.4.3-1 may be used for selection of δ if other data are not available. For long piles f may not indefinitely increase linearly with the overburden pressure as implied by Eq. 6.4.3-1. In such cases it may be appropriate to limit f to the values given in Table 6.4.3-1.

For piles end bearing in cohesionless soils the unit end bearing q in lb/ft² (kPa) may be computed by the equation

$$q = p_o N_q \quad (6.4.3-2)$$

where

$p_o =$ effective overburden pressure lb/ft² (kPa) at the pile tip,

$N_q =$ dimensionless bearing capacity factor.

Table 6.4.3-1—Design Parameters for Cohesionless Siliceous Soil*

Density	Soil Description	Soil-Pile Friction Angle, δ Degrees	Limiting Skin Friction Values kips/ft ² (kPa)	N_q	Limiting Unit End Bearing Values kips/ft ² (MPa)
Very Loose	Sand	15	1.0 (47.8)	8	40 (1.9)
Loose	Sand-Silt**				
Medium	Silt				
Loose	Sand	20	1.4 (67.0)	12	60 (2.9)
Medium	Sand-Silt**				
Dense	Silt				
Medium	Sand	25	1.7 (81.3)	20	100 (4.8)
Dense	Sand-Silt**				
Dense	Sand	30	2.0 (95.7)	40	200 (9.6)
Very Dense	Sand-Silt**				
Dense	Gravel	35	2.4 (114.8)	50	250 (12.0)
Very Dense	Sand				

*The parameters listed in this table are intended as guidelines only. Where detailed information such as in situ cone tests, strength tests on high quality samples, model tests, or pile driving performance is available, other values may be justified.

**Sand-Silt includes those soils with significant fractions of both sand and silt. Strength values generally increase with increasing sand fractions and decrease with increasing silt fractions.

Recommended values of N_q are presented in Table 6.4.3-1. The shaft friction, f , acts on both the inside and outside of the piles. However, the total resistance in excess of the external shaft friction plus annular end bearing is the total internal shaft friction or the end bearing of the plug, whichever is less. For piles considered to be plugged the bearing pressure may be assumed to act over the entire cross section of the pile. For unplugged piles the bearing pressure acts on the pile annulus only. Whether a pile is considered to be plugged or unplugged may be based on static calculations. For example, a pile could be driven in an unplugged condition but act plugged under static loading.

Load test data for piles in sand (see “Comparison of Measured and Axial Load Capacities of Steel Pipe Piles in Sand with Capacities Calculated Using the 1986 API RP2A Standard,” Final Report to API, Dec. 1987, by R. E. Olson) indicate that variability in capacity predictions may exceed those for piles in clay. Other data (see Toolan and Ims (1988)) (1) suggest that for piles in loose sands and long piles (> 50 m) in tension the method may be less conservative than for compression piles in medium dense to dense sands. Therefore, in unfamiliar situations, the designer may want to account for

this uncertainty through a selection of conservative design parameters and/or safety factors. This may be especially important where load shedding subsequent to peak load development leading to an abrupt (brittle) failure may occur such as the case for short piles under tension loading.

For soils that do not fall within the ranges of soil density and description given in Table 6.4.3-1, or for materials with unusually weak grains or compressible structures, Table 6.4.3-1 may not be appropriate for selection of design parameters. For example, very loose silts or soils containing large amounts of mica or volcanic grains may require special laboratory or field tests for selection of design parameters. Of particular importance are sands containing calcium carbonate which are found extensively in many areas of the oceans. Available data suggest that driven piles in these soils may have substantially lower design strength parameters than given in Table 6.4.3-1. Drilled and grouted piles in carbonate sands, however, may have significantly higher capacities than driven piles and have been used successfully in many carbonate areas. The characteristics of carbonate sands are highly variable and local experience should dictate the design parameters selected. For example, available qualitative data

suggest that capacity is improved in carbonate soils of high densities and higher quartz contents. Cementation may increase end bearing capacity, but result in a loss of lateral pressure and a corresponding decrease in frictional capacity. These materials are discussed further in the Commentary.

For piles driven in undersized drilled or jetted holes in cohesionless soils the values of f and q should be determined by some reliable method that accounts for the amount of soil disturbance due to installation but they should not exceed values for driven piles. Except in unusual soil types such as described above, the values of f and q in Table 6.4.3-1 may be used for drilled and grouted piles, with consideration given to the strength of the soil grout interface.

In layered soils shaft friction values, f , in the cohesionless layers should be as outlined in Table 6.4.3-1. End bearing values for piles tipped in cohesionless layers with adjacent soft layers may also be taken from Table 6.4.3-1, assuming that the pile achieves penetration of two to three diameters or more into the cohesionless layer, and the tip is approximately three diameters above the bottom of the layer to preclude punch through. Where these distances are not achieved, some modification in the tabulated values may be necessary. Where adjacent layers are of comparable strength to the layer of interest, the proximity of the pile tip to the interface is not a concern.

6.4.4 Skin Friction and End Bearing of Grouted Piles in Rock

The unit skin friction of grouted piles in jetted or drilled holes in rock should not exceed the triaxial shear strength of the rock or grout, but in general should be much less than this value based on the amount of reduced shear strength from installation. For example the strength of dry compacted shale may be greatly reduced when exposed to water from jetting or drilling. The sidewall of the hole may develop a layer of slaked mud or clay which will never regain the strength of the rock. The limiting value for this type pile may be the allowable bond stress between the pile steel and the grout as recommended in 7.4.3.

The end bearing capacity of the rock should be determined from the triaxial shear strength of the rock and an appropriate bearing capacity factor based on sound engineering practice for the rock materials but should not exceed 100 tons per square foot (9.58 MPa).

6.5 PILE CAPACITY FOR AXIAL PULLOUT LOADS

The ultimate pile pullout capacity may be equal to or less than but should not exceed Q_f , the total skin friction resistance. The effective weight of the pile including hydrostatic uplift and the soil plug shall be considered in the analysis to determine the ultimate pullout capacity. For clay, f should be the same as stated in 6.4.2. For sand and silt, f should be computed according to 6.4.3.

For rock, f should be the same as stated in Section 6.4.4.

The allowable pullout capacity should be determined by applying the factors of safety in 6.3.4 to the ultimate pullout capacity.

6.6 AXIAL PILE PERFORMANCE

6.6.1 Static Load-Deflection Behavior

Piling axial deflections should be within acceptable serviceability limits and these deflections should be compatible with the structural forces and movements. An analytical method for determining axial pile performance is provided in *Computer Predictions of Axially Loaded Piles with Non-linear Supports*, by P. T. Meyer, et al., OTC 2186, May 1975. This method makes use of axial pile shear transition vs. local pile deflection (t - z) curves to model the axial support provided by the soil along the size of the pile. An additional (Q - z) curve is used to model the tip and bearing vs. the deflection response. Methods for constructing t - z and Q - z curves are given in Section 6.7. Pile response is affected by load directions, load types, load rates, loading sequence installation technique, soil type, axial pile stiffness and other parameters.

Some of these effects for cohesive soils have been observed in both laboratory and field tests.

In some circumstances, i.e., for soils that exhibit strain-softening behavior and/or where the piles are axially flexible, the actual capacity of the pile may be less than that given by Eq. 6.4.1-1. In these cases an explicit consideration of these effects on ultimate axial capacity may be warranted. Note that other factors such as increased axial capacity under loading rates associated with storm waves may counteract the above effects. For more information see Section 6.2.2, its commentary, as well as "Effects of Cyclic Loading and Pile Flexibility on Axial Pile Capacities in Clay" by T. W. Dunnivant, E. C. Clukey and J. D. Murff, OTC 6374, May 1990.

6.6.2 Cyclic Response

Unusual pile loading conditions or limitations on design pile penetrations may warrant detailed consideration of cyclic loading effects.

Cyclic loadings (including inertial loadings) developed by environmental conditions such as storm waves and earthquakes can have two potentially counteractive effects on the static axial capacity. Repetitive loadings can cause a temporary or permanent decrease in load-carrying resistance, and/or an accumulation of deformation. Rapidly applied loadings can cause an increase in load-carrying resistance and/or stiffness of the pile. Very slowly applied loadings can cause a decrease in load-carrying resistance and/or stiffness of the pile. The resultant influence of cyclic loadings will be a function of the combined effects of the magnitudes, cycles, and rates of applied pile loads, the structural characteristics of the

pile, the types of soils, and the factors of safety used in design of the piles.

The design pile penetration should be sufficient to develop an effective pile capacity to resist the design static and cyclic loadings as discussed in 6.3.4.

The design pile penetration can be confirmed by performing pile response analyses of the pile-soil system subjected to static and cyclic loadings. Analytical methods to perform such analyses are described in the commentary to this Section. The pile-soil resistance-displacement t - z , Q - z characterizations are discussed in Section 6.7.

6.6.3 Overall Pile Response Analyses

When any of the above effects are explicitly considered in pile response analysis, the design static and cyclic loadings should be imposed on the pile top and the resistance-displacements of the pile determined. At the completion of the design loadings, the maximum pile resistance and displacement should be determined. Pile deformations should meet structure serviceability requirements. The total pile resistance after the design loadings should meet the requirements of 6.3.4.

6.7 SOIL REACTION FOR AXIALLY LOADED PILES

6.7.1 General

The pile foundation should be designed to resist the static and cyclic axial loads. The axial resistance of the soil is provided by a combination of axial soil-pile adhesion or load transfer along the sides of the pile and end bearing resistance at the pile tip. The plotted relationship between mobilized soil-pile shear transfer and local pile deflection at any depth is described using a t - z curve. Similarly, the relationship between mobilized end bearing resistance and axial tip deflection is described using a Q - z curve.

6.7.2 Axial Load Transfer (t - z) Curves

Various empirical and theoretical methods are available for developing curves for axial load transfer and pile displacement, (t - z) curves. Theoretical curves described by Kraft, et al. (1981) may be constructed. Empirical t - z curves based on the results of model and full-scale pile load tests may follow the procedures in clay soils described by Cole and Reese (1966) or granular soils by Coyle, H.M. and Suliaman, I.H. *Skin Friction for Steel Piles in Sand*, Journal of the Soil Mechanics and Foundation Division, Proceedings of the American Society of Civil Engineers, Vol. 93, No. SM6, November, 1967, p. 261–278. Additional curves for clays and sands are provided by Vijayvergiya, V.N., *Load Movement Characteristics of Piles*, Proceedings of the Ports '77 Conference, American Society of Civil Engineers, Vol. II, p. 269–284.

Load deflection relationships for grouted piles are discussed in *Criteria for Design of Axially Loaded Drilled Shafts*, by L. C. Reese and M. O'Neill, Center for Highway Research Report, University of Texas, August 1971. Curves developed from pile load tests in representative soil profiles or based on laboratory soil tests that model pile installation may also be justified. Other information may be used, provided such information can be shown to result in adequate safeguards against excessive deflection and rotation.

In the absence of more definitive criteria, the following t - z curves are recommended for non-carbonate soils. The recommended curves are shown in Figure 6.7.2-1.

Clays	t/D	t/t_{max}
	0.0016	0.30
	0.0031	0.50
	0.0057	0.75
	0.0080	0.90
	0.0100	1.00
	0.0200	0.70 to 0.90
	∞	0.70 to 0.90
Sands	t (in.)	t/t_{max}
	0.000	0.00
	0.100	1.00
	∞	1.00

where

t = local pile deflection, in. (mm),

D = pile diameter, in. (mm),

t = mobilized soil pile adhesion, lb/ft² (kPa),

t_{max} = maximum soil pile adhesion or unit skin friction capacity computed according to Section 6.4, lb/ft² (kPa).

The shape of the t - z curve at displacements greater than z_{max} as shown in Figure 6.7.2-1 should be carefully considered. Values of the residual adhesion ratio t_{res}/t_{max} at the axial pile displacement at which it occurs (z_{res}) are a function of soil stress-strain behavior, stress history, pipe installation method, pile load sequence and other factors.

The value of t_{res}/t_{max} can range from 0.70 to 0.90. Laboratory, in situ or model pile tests can provide valuable information for determining values of t_{res}/t_{max} and z_{res} for various soils. For additional information see the listed references at the beginning of 6.7.2.

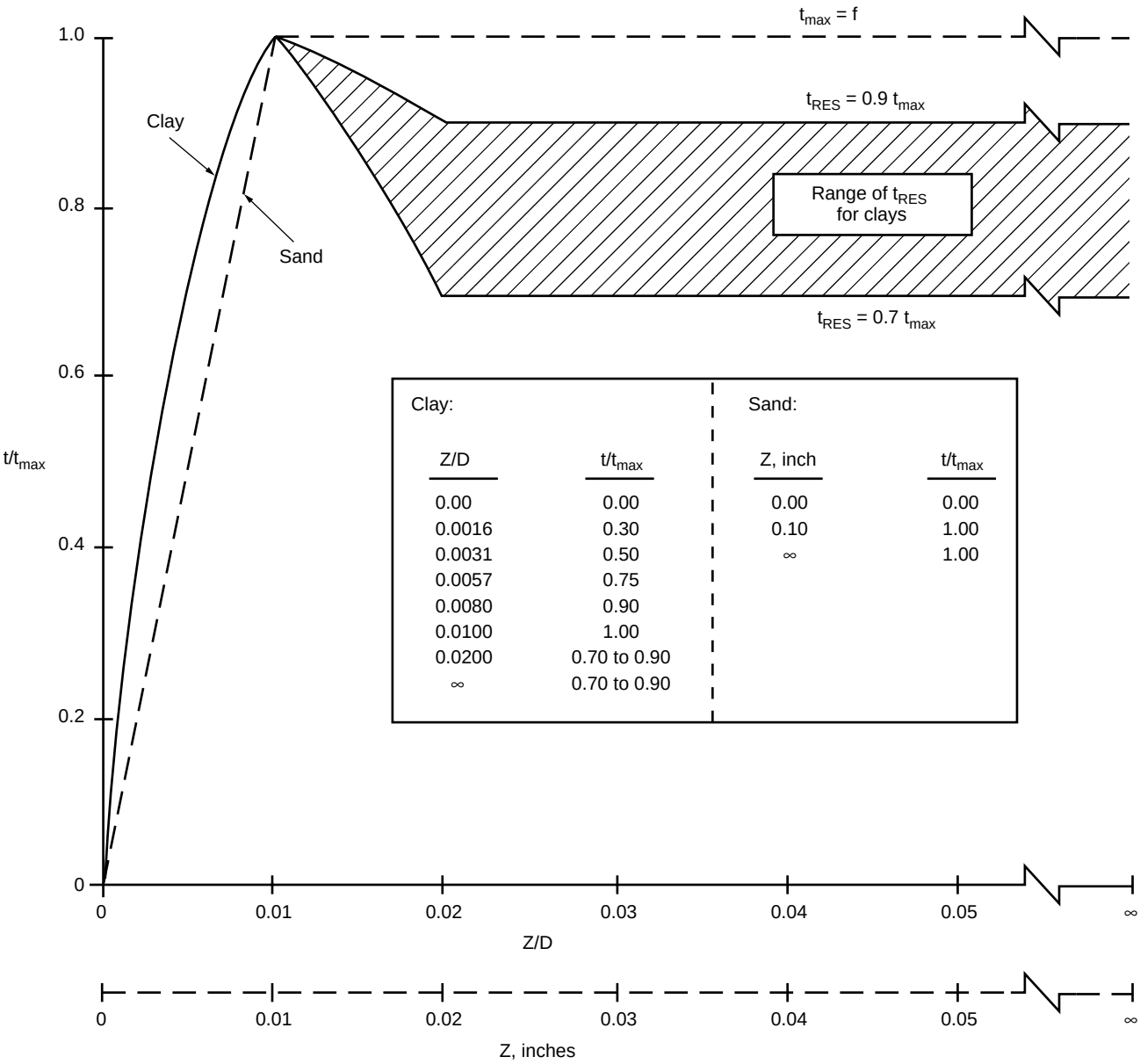


Figure 6.7.2-1—Typical Axial Pile Load Transfer-Displacement (t - z) Curves

6.7.3 Tip Load—Displacement Curve

The end bearing or tip load capacity should be determined as described in 6.4.2 and 6.4.3. However, relatively large pile tip movements are required to mobilize the full end bearing resistance. A pile tip displacement up to 10 percent of the pile diameter may be required for full mobilization in both sand and clay soils. In the absence of more definitive criteria the following curve is recommended for both sands and clays.

z/D	Q/Q_p
0.002	0.25
0.013	0.50
0.042	0.75
0.073	0.90
0.100	1.00

where

z = axial tip deflection, in. (mm),

D = pile diameter, in. (mm),

Q = mobilized end bearing capacity, lb (kN).

Q_p = total end bearing, lb (kN), computed according to Section 6.4.

The recommended curve is shown in Figure 6.7.3-1.

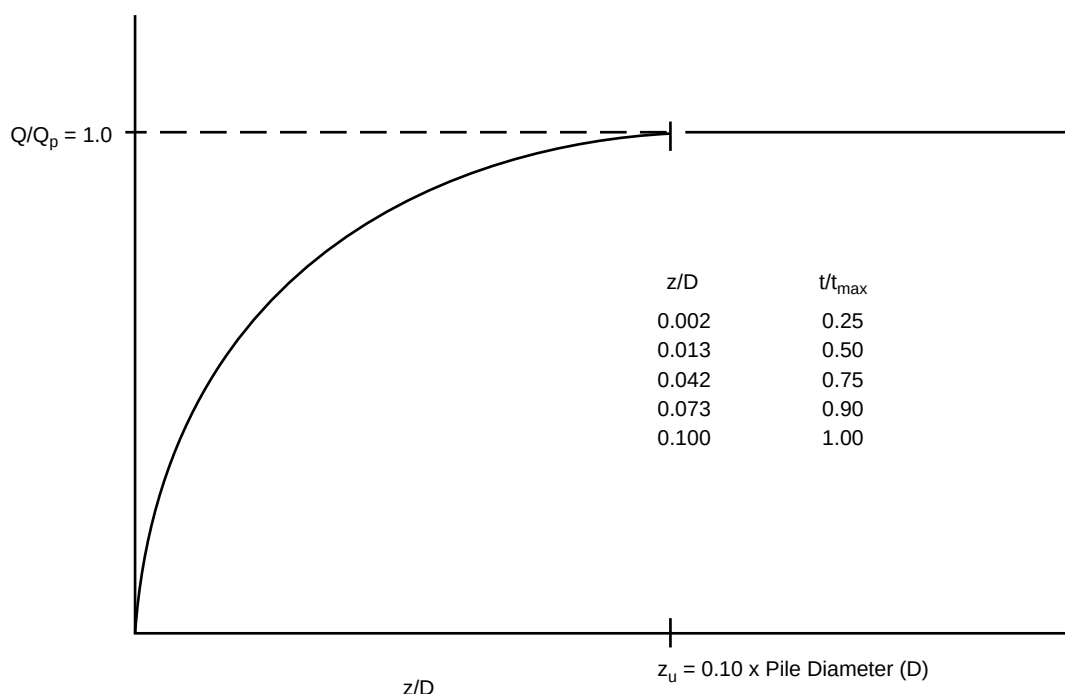


Figure 6.7.3-1—Pile Tip-Load-Displacement (Q - z) curve

6.8 SOIL REACTION FOR Laterally-LOADED PILES

6.8.1 General

The pile foundation should be designed to sustain lateral loads, whether static or cyclic. Additionally, the designer should consider overload cases in which the design lateral loads on the platform foundation are increased by an appropriate safety factor. The designer should satisfy himself that the overall structural foundation system will not fail under the overloads. The lateral resistance of the soil near the surface is significant to pile design, and the effects on this resistance of scour and soil disturbance during pile installation should be considered. Generally, under lateral loading, clay soils behave as a plastic material which makes it necessary to relate pile-soil deformation to soil resistance. To facilitate this procedure, lateral soil resistance deflection (p - y) curves should be constructed using stress-strain data from laboratory soil samples. The ordinate for these curves is soil resistance, p , and the abscissa is soil deflection, y . By iterative procedures, a compatible set of load-deflection values for the pile-soil system can be developed.

For a more detailed study of the construction of p - y curves refer to the following publications.

Soft Clay: OTC 1204, Correlations for Design of Laterally Loaded Piles in Soft Clay, by H. Matlock, April 1970.